



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Approximate Multiplier Unit with Reversible Logic for RISC-V Architecture

Mentors: Dr.Gaganpreet Kaur & Dr.Mohit Agarwal

Team Members:

Aryan Sadana (102306041)

Archit Mittal (102322030)

Aviral Arora (102322059)

Rudraa Dadhich (102322068)

Bani Singh (102322051)

Current Progress



4-bit Wallace tree

Architecture designed



Reversible gate optimization

Replaced HA/FA in tree



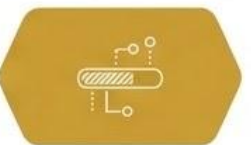
Testing & verification

Approx. multiplier verified



Scaled to 8-bit tree

Architecture scaled up

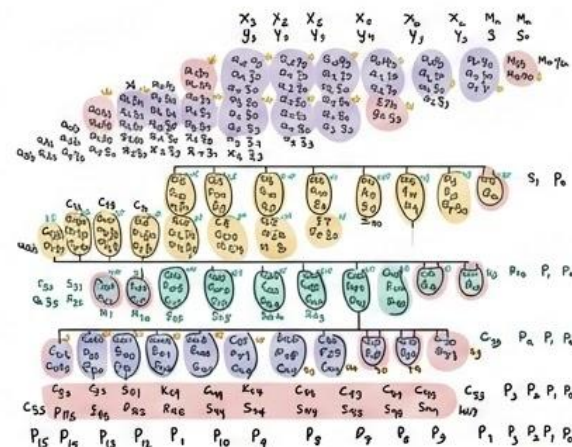
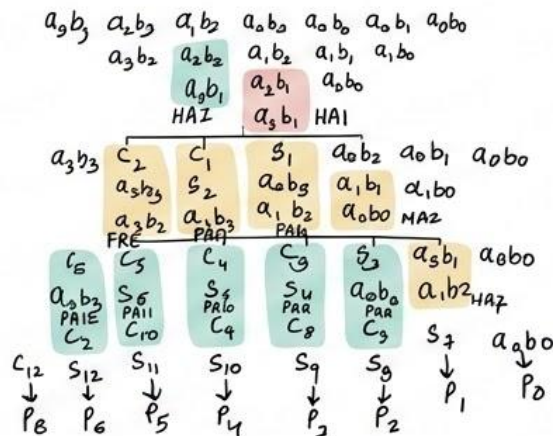


8-bit optimization

In progress: gate search

Completed

In progress



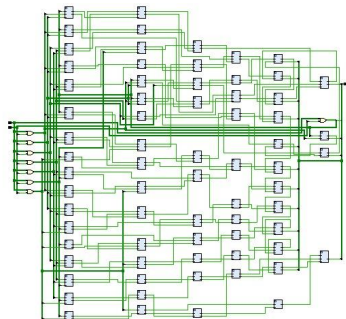
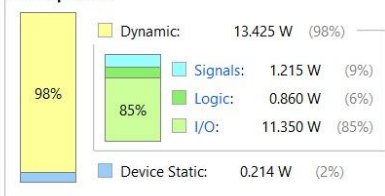
Results

- Best 8-bit approximate multiplier results obtained so far are from **Vivado implementation** (post-synthesis/post-implementation); **on-FPGA validation is pending**
- Observed trend: **increasing gate count does not proportionally reduce power** - indicating diminishing returns from further approximation at this stage
- Additional constraints identified: **thermal warnings** during implementation, alongside **accuracy degradation (error)** that must be balanced against power savings

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power: 13.638 W (Junction temp exceeded!)
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 93.2°C

On-Chip Power



Destination: (input port) F[14]
 Path Group: (output port) (none)
 Path Type: Max at Slow Process Corner
 Data Path Delay: 10.802ns (logic 4.400ns)
 Logic Levels: 9 (IBUF=1 LUT3=1 LUT6=6 O
 Location Delay type I

Design Configuration	Gate Units	Power (Static)	Power (Dynamic)	Total Power	Time (ns)
Original design (no reversible gates)	0	0.221	13.687	13.908	10.560
2FA Stage 1 (TSG)	2	0.221	13.697	13.918*	10.401
1HA Stage 1 (DPG) + 2FA (TSG)	3	0.221	13.697	13.918*	10.401
1HA + 2FA (Stage 1, all DPG)	3	0.214	13.458	13.673	10.360
Stage 1 (2HA 4FA) + Stage 2 (1HA 1FA) (all DPG)	8	0.214	13.425	13.638	10.802
Stage1 1HA (DKG) 2FA (DPG) Stage 2 1HA (DKG) 1FA (DPG)	5	0.214	13.425	13.638	10.802
Stage 1 [2HA (DPG) 2FA (DPG)] + Stage 2 [1HA + 1FA (DPG)]	6	0.214	13.425	13.638	10.802